Two medial orbits require two embedded intersection tests

Human Compression project

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Abstract

We construct a finite four-polytope in class $2_{\{1,2\}}$ with exactly two orbits of medial sections. Its smallest regular cover is polytopal, but the intersection condition printed in Cunningham–Hubard, arXiv:2506.19334v1, Corollaries 6.10 and 6.15(2), is false. The example has 144 flags; its regular cover has 2592 flags. The distinction is between subgroups embedded at different paired roots, rather than between abstract section types. The proof establishes all polytopality and automorphism assertions. A complete finite replay also checks the actual graphs and subgroup embeddings.

1 A finite two-orbit polytope

All coordinates lie in $\mathbb{F}_3$. Let D be the affine group of the line generated by $s(z) = -z$ and $t(z) = 1-z$. Their product is a nonidentity translation, so D is dihedral of order six. On $\mathbb{F}_3^2$ put

$$a(x,y) = (-x,y), \quad b(x,y) = (x,1-y), \quad d(x,y) = (y,x), \quad a' = dad, \quad b' = dbd. \tag{1}$$

Composition here is ordinary function composition. The group $H = \langle a,b,d\rangle$ is $(D\times D)\rtimes\langle d\rangle$, of order 72: a, a', b, b' generate the two copies of D, and d interchanges them.

Define a colored graph P on the 144 pairs (h,ϵ), with $h \in H$ and $\epsilon \in \{0,1\}$, by

$$\begin{array}{ll} r_0(h,\epsilon) = (h,1-\epsilon), & r_1(h,\epsilon) = (a_\epsilon h,\epsilon), \\ r_2(h,\epsilon) = (bh,\epsilon), & r_3(h,\epsilon) = (dh,1-\epsilon), \end{array} \qquad a_0 = a, \quad a_1 = a'. \tag{2}$$

These are fixed-point-free involutions with distinct neighbors at each flag. Nonadjacent colors commute; the only case depending on (1) is $a_{1-\epsilon}d = da_\epsilon$. The graph is connected: colors $0,3$ supply multiplication by d without changing ϵ, while at $\epsilon = 0$ the other colors supply a, b. Thus (2) is a maniplex.

We use the standard path-intersection characterization of polytopality: at every flag, orbits under two color sets intersect in the orbit under their intersection. Its recursive form says that polytopality of all facets and vertex-figures, together with their exact intersection at each flag, suffices. The companion common paper proves this recursive form from the established maniplex–polytope correspondence. It applies to digons as well as to polygons with more sides.

Lemma 1. *The graph P is a four-polytope. Its full color-preserving automorphism group is the right regular action of H. It has class $2_{\{1,2\}}$, exactly two medial orbits, and one orbit each of facets and vertex-figures. Its medial sections are digons and triangles.*

Proof. Right multiplication $R_k(h,\epsilon) = (hk,\epsilon)$ commutes with all adjacencies and is transitive on each set defined by ϵ. Thus it suffices to check local intersections at $(1,0)$ and $(1,1)$. Products below refer to the two coordinate actions on $\mathbb{F}_3^2$. The facet through either base flag has the same H-coordinates in both sets, namely $A_H = \langle a\rangle \times D$. The vertex-figure through $(1,0)$ has coordinates $B_H = D \times \langle b\rangle$ at $\epsilon = 0$ and dB_H at $\epsilon = 1$. Through $(1,1)$ these are $B'_H = \langle b'\rangle \times D$ at $\epsilon = 1$ and dB'_H at $\epsilon = 0$. Eliminating the color which changes ϵ proves these descriptions; for example the even words in colors $1,2,3$ generate

a, b, b' in the first case. Diagonal coordinate actions and actions which interchange coordinates are disjoint. Moreover,

$$A_H \cap B_H = \langle a, b \rangle, \qquad A_H \cap B'_H = \langle a', b \rangle. \tag{3}$$

The distinct reflections s, t have trivial subgroup intersection on the first coordinate. The two right sides in (3) have orders four and six, respectively. This gives the required top intersection at both flags.

For the facet check, the even part of the colors $0, 1$ orbit is $\langle a, a' \rangle$, and the odd part has the same coordinates. Its intersections with the two medial groups are $\langle a \rangle$ and $\langle a' \rangle$, respectively. For the vertex-figure check, the even part of the colors $2, 3$ orbit is $\langle b, b' \rangle$, and the odd part is its left translate by d. Its intersection with either medial group is $\langle b \rangle$. These assertions again use $\langle s \rangle \cap \langle t \rangle = 1$. All rank-two residues are alternating cycles and therefore polygons. The recursive criterion proves both kinds of rank-three residues and then, by (3), the whole polytope.

An automorphism of a connected colored graph is determined by the image of one flag. One interchanging the two sets defined by ϵ would send a middle-color component of size four to one of size six, which is impossible. The right action of H is already regular on each set, so it is the full automorphism group. Colors $1, 2$ preserve its orbits; colors $0, 3$ interchange them. Every medial component stays in one set, and H is transitive there: the two different sizes give precisely two medial orbits. Each facet and vertex-figure meets both sets, so H is transitive on those faces. In particular all facets are isomorphic and all vertex-figures are isomorphic, as the source requires. □

2 The regular cover and its fibers

Let $\overline{P}$ denote P rooted at $(1, 1)$ instead of $(1, 0)$. The connected mix $Q = P \diamond \overline{P}$ starts at

$$q_0 = ((1, 0), (1, 1)), \qquad q_1 = r_0 q_0 = ((1, 1), (1, 0)).$$

Write its flags as (h_0, h_1, ϵ), representing $((h_0, \epsilon), (h_1, 1 - \epsilon))$. Those with $\epsilon = 0$ are exactly the pairs in

$$K = \langle (a, a'), (b, b), (d, d) \rangle \leq H \times H. \tag{4}$$

The displayed generators are realized by colors 1, 2, and $0, 3$. Conversely, eliminate color 0 from an even word; its conjugation sends (a, a') to (a', a), already the conjugate by (d, d), and fixes the other two generators. Flags at $\epsilon = 1$ have the same pairs, since color 0 changes only ϵ. Right multiplication by K acts regularly on each set of flags. Interchanging the two input flags sends q_0 to q_1, preserves the connected component, and commutes with all colors. Consequently Q is regular.

For completeness $|K| = 1296$. The squares of products of (b, b) with (a, a') and with (a', a) give independent translations in the two y-coordinates. Conjugation by (d, d) gives both x-translations. Hence K contains all 3^4 translations. Its linear image has the three independent sign vectors

$$(1, 0, 0, 1), \quad (0, 1, 1, 0), \quad (0, 1, 0, 1)$$

and simultaneous interchange of the two axes in both factors. The coordinate order here is x_0, y_0, x_1, y_1; a 1 denotes a sign change. The span is stable under the interchange, which has order two outside the sign subgroup. Thus the linear image has order 16, and Q has $2 \cdot 3^4 \cdot 16 = 2592$ flags.

For a color set J and $q_k = (x_k, y_k)$ define the actual fiber

$$F_J(q_k) = \{h \in H : (x_k, R_h y_k) \text{ lies in the } J\text{-component of } q_k\}.$$

These are the embedded lower variance subgroups in the second input's automorphism group. The right regular action identifies this group with H^{op}, which changes none of its subgroups or intersections. Set

$A_k = F_{\{0,1,2\}}(q_k)$, $B_k = F_{\{1,2,3\}}(q_k)$ and $C_k = F_{\{1,2\}}(q_k)$. For $u \in \{0,1\}$ write $\eta = (-1)^u$. Inside this single specified copy of $\mathrm{Aut}(P)$ we have

$$\begin{aligned} A_0 &= A_1 = \{(x,y) \mapsto (\eta x, \eta y + c) : u = 0,1,\ c \in \mathbb{F}_3\}, \\ B_0 &= \{(x,y) \mapsto (\eta x + u, \eta y + c) : u = 0,1,\ c \in \mathbb{F}_3\}, \\ B_1 &= \{(x,y) \mapsto (\eta x + c, \eta y + u) : u = 0,1,\ c \in \mathbb{F}_3\}, \\ C_0 &= \{(x,y) \mapsto (x, y + c) : c \in \mathbb{F}_3\}, \qquad C_1 = \langle (x,y) \mapsto (-x, 1-y) \rangle. \end{aligned} \tag{5}$$

Here are complete generation checks. Put $U = (a, a')$, $W = (a', a)$, $V = (b, b)$ and $V' = (b', b')$. The even part of a mixed facet is $\langle U, W, V \rangle$, consisting exactly of the pairs

$$\begin{aligned} h_0(x,y) &= ((-1)^u x,\ (-1)^{w+v} y + \alpha), \\ h_1(x,y) &= ((-1)^w x,\ (-1)^{u+v} y + \beta), \end{aligned} \tag{6}$$

where $u, w, v \in \{0,1\}$ and $\alpha, \beta \in \mathbb{F}_3$. The generators give the independent signs, while $(UV)^2$ and $(WV)^2$ give separate nonzero translations on the two y-coordinates. Thus every displayed pair occurs. Setting $h_0 = 1$ gives A_0; factor interchange gives the same set for A_1.

The even part of the mixed vertex-figure at q_0 is $\langle U, V, V' \rangle$. Its elements are precisely

$$\begin{aligned} h_0(x,y) &= ((-1)^{u+w} x + \alpha,\ (-1)^v y + v), \\ h_1(x,y) &= ((-1)^w x + w,\ (-1)^{u+v} y + \beta). \end{aligned} \tag{7}$$

The signs are independent. The squares $(UV)^2$ and $(UV')^2$ give independent translations on y_1 and x_0. For each remaining coordinate the map is z or $1 - z$, so its translation is forced by its sign; these conditions are closed under composition. This proves equality in (7). Setting $h_0 = 1$ yields B_0, and setting $h_1 = 1$ yields B_1 at the reversed paired root. Finally the middle group $\langle U, V \rangle$ is dihedral of order twelve: its product has order six, with first projection ab of order two and second projection $a'b$ of order three. Its projection kernels have orders three and two and give exactly the two groups C_0, C_1. Consequently

$$A_0 \cap B_0 = C_0 \cong C_3, \qquad A_1 \cap B_1 = C_1 \cong C_2, \qquad C_0 \cap C_1 = 1. \tag{8}$$

Theorem 2. *The smallest regular cover of P is polytopal. Nevertheless $P \diamond \overline{P}$ violates the condition printed in [CH, Corollary 6.10(3)] and the medial variance condition printed in [CH, Corollary 6.15(2)]. All exceptional two-orbit hypotheses hold.*

Proof. Every facet and vertex-figure of Q is a mix of two polyhedra and is polytopal. Here the usual polyhedron-mix result also has a short path proof. An intersection of the two rank-three end orbits has its second-coordinate fiber inside one middle edge of the second polyhedron. Its nonidentity endpoint would require a word fixing a polygon flag in the first factor but changing polygon flag parity in the second. Polygon flag graphs are bipartite, so this is impossible. The common fiber-intersection lemma proves the rank-three intersection property, including when a polygon is a digon.

At q_0, (8) is exactly the top fiber-intersection condition. Regularity of Q transports the resulting path condition to every flag, whether or not that automorphism projects to automorphisms of the individual rooted factors. The recursive criterion therefore proves polytopality of Q.

Let W be the universal string connection group and N the stabilizer of $(1,0)$ in P. Its index-two subgroup V preserving ϵ normalizes N: on each of its two orbits the right group H acts regularly and centralizes V, so the image of V acts regularly as well. Thus the core of N in W is $N \cap r_0 N r_0$, the stabilizer of q_0 in the mix. It follows that Q is the smallest regular premaniplex cover. Since it is polytopal, it is also the smallest regular polytopal cover.

Lemma 1 shows that both inputs genuinely belong to $2_{\{1,2\}}$, with uniform facet and vertex-figure types and precisely two medial orbits. Corollary 6.10(3) would require $A_0 \cap B_0 \leq C_0 \cap C_1$. Formula (8) contradicts it: the left side has order three and the right side is trivial. A literal witness is the color

sequence $1, 2, 1, 2$ at q_0. It fixes the first flag and moves the second by $(x, y) \mapsto (x, y+2)$, a nonidentity member of both end fibers outside C_1.

For Corollary 6.15(2), the two medial sections are a digon and a triangle. Their two lower variance groups have orders three and two, so their intersection is trivial under any embeddings in one ambient group. The printed order $X(K \mid \overline{K})$ puts P second, so use the reversed factor order $\overline{P} \diamond P$: in our single ambient copy of H this is the calculation at q_1, with common end variance group $A_1 \cap B_1 = C_1$ of order two. Both end mixes are polytopal, but this nontrivial group cannot lie in the stated trivial intersection. Reversing the initial base flag gives instead order three on the left, with the same failure. No regular-input degeneration or identification of abstract group types is being used. □

3 Corrected criterion and scope

The valid exceptional criterion keeps the two actual paired roots:

$$A_0 \cap B_0 \leq C_0, \qquad A_0 \cap B_1 \leq C_1. \tag{9}$$

The equality $A_0 = A_1$ holds because the flags lie in one mixed facet. Their vertex-figures are isomorphic but are different faces, giving no equality of the embedded stabilizers B_0, B_1. Even the weaker proposed equality $A_0 \cap B_0 = A_0 \cap B_1$ fails, by the different orders in (8). The companion paper proves (9) in general. For the smallest regular cover, regularity permits one correctly rooted test; it does not permit intersecting two differently transported medial subgroups on its right.

This addition resolves the two qualified entries in our frozen F005 recovery packet without changing that packet. It does not answer the three final problems of [CH] or assert a compression factor or historical priority. The universal connection-group, path-intersection and regular-cover correspondences are retained standard inputs. The affine construction and all the intersections above are proved here. The complete program is a validation of these written proofs, not a premise replacing a classification or geometric argument.

References

[1] G. Cunningham and I. Hubard, *Polytopality criteria for the mix of polytopes and maniplexes*, arXiv:2506.19334v1, 24 June 2025, physical pp. 17 and 19, Corollaries 6.10 and 6.15(2).

[2] Human Compression project, *Intersections in simultaneous group actions* and *Polytopality of mixes and their regular covers*, frozen author packet `topic_Mix_F005_recovery`, 2026; rooted exceptional tests and their exact common-facet equality.